

Environment Monitoring Controller using ESP32 and Firebase (Simulation Project)

1.Abstract

This project presents the design and development of an IoT-based real-time data monitoring system using the ESP32 microcontroller and Firebase Realtime Database. The main objective of this system is to collect environmental sensor data, transmit it to the cloud, and enable remote monitoring through a web interface. The system is designed to demonstrate how embedded hardware and cloud computing can be integrated to build efficient and scalable IoT applications. In this project, the ESP32 acts as the central processing unit due to its built-in WiFi capability, which makes it highly suitable for Internet of Things applications. The system generates or reads sensor values such as temperature and humidity. These values are processed by the ESP32 and then sent to the Firebase cloud using an HTTP REST API. Firebase Realtime Database is used as the cloud platform because it provides fast data storage and real-time synchronization across devices. Once the data is uploaded to Firebase, it can be accessed from anywhere using a web browser. A simple web dashboard is created using HTML and JavaScript to display the live sensor values. The system continuously updates the data at regular intervals, ensuring real-time monitoring without delay. The proposed system is cost-effective, efficient, and easy to implement. It eliminates the need for manual data collection and enables remote access to sensor information. This type of system is highly useful in various applications such as smart home automation, environmental monitoring systems, industrial monitoring, agriculture systems, and smart city projects.

Overall, this project helps in understanding the fundamentals of IoT communication, cloud integration, and real-time data visualization. It also provides practical experience in working with ESP32, Firebase, and web technologies, making it a strong foundation for advanced IoT-based projects in the future.

2.Introduction

The Internet of Things (IoT) is a rapidly growing technology that enables physical devices to connect and communicate with the internet. In this project, an **ESP32 microcontroller** is used as the main controller because it has built-in WiFi capability, making it suitable for IoT applications. The system reads sensor values (temperature and humidity) and sends the data to the **Firebase Realtime Database**, which acts as a cloud storage platform. The data can then be accessed and displayed on a web page in real time. This project helps to understand the basic concept of IoT communication, cloud data storage, and real-time monitoring. It also provides hands-on experience in using ESP32, Firebase, and web technologies together in a single system.

3.Components Used

- ESP32 (Wokwi Simulator)
- Firebase Realtime Database
- HTTP REST API
- WiFi (Wokwi-GUEST)

4.Working Principle

This project is a real-time IoT environmental monitoring system using ESP32 and Firebase Realtime Database. It collects temperature and humidity data, sends it to the cloud, and displays it in real time through a web browser.

i. Sensor Data Generation (Simulation)

In this project, physical sensors are not used. Instead, Wokwi simulation is used to generate virtual sensor values. Temperature and humidity values are created using a program (random function or predefined logic), representing real environmental conditions.

ii. Data Acquisition by ESP32

The ESP32 microcontroller acts as the main controller. It continuously reads the simulated sensor values and processes them. The collected data is then structured in JSON format for easy transmission.

Example:

```
{  
  "temperature": 26,  
  "humidity": 61  
}
```

iii. Wi-Fi Connectivity

The ESP32 connects to the internet using the Wokwi-GUEST Wi-Fi network in the simulation environment. This enables communication between the device and the cloud server.

iv. Data Transmission using HTTP REST API

The ESP32 sends the sensor data to Firebase Realtime Database using HTTP REST API (PUT/POST request). Here, ESP32 acts as a client, and Firebase acts as the cloud server. This allows secure transmission of data over the internet.

v. Firebase Realtime Database (Cloud Storage)

Firebase stores the received data and updates it instantly in real time. Only the latest temperature and humidity values are maintained in the database, ensuring live monitoring.

vi. Data Visualization

The stored data is accessed through a web browser. Users can view live temperature and humidity values, which update automatically without refreshing the page due to Firebase's real-time synchronization feature.

5. Block Diagram

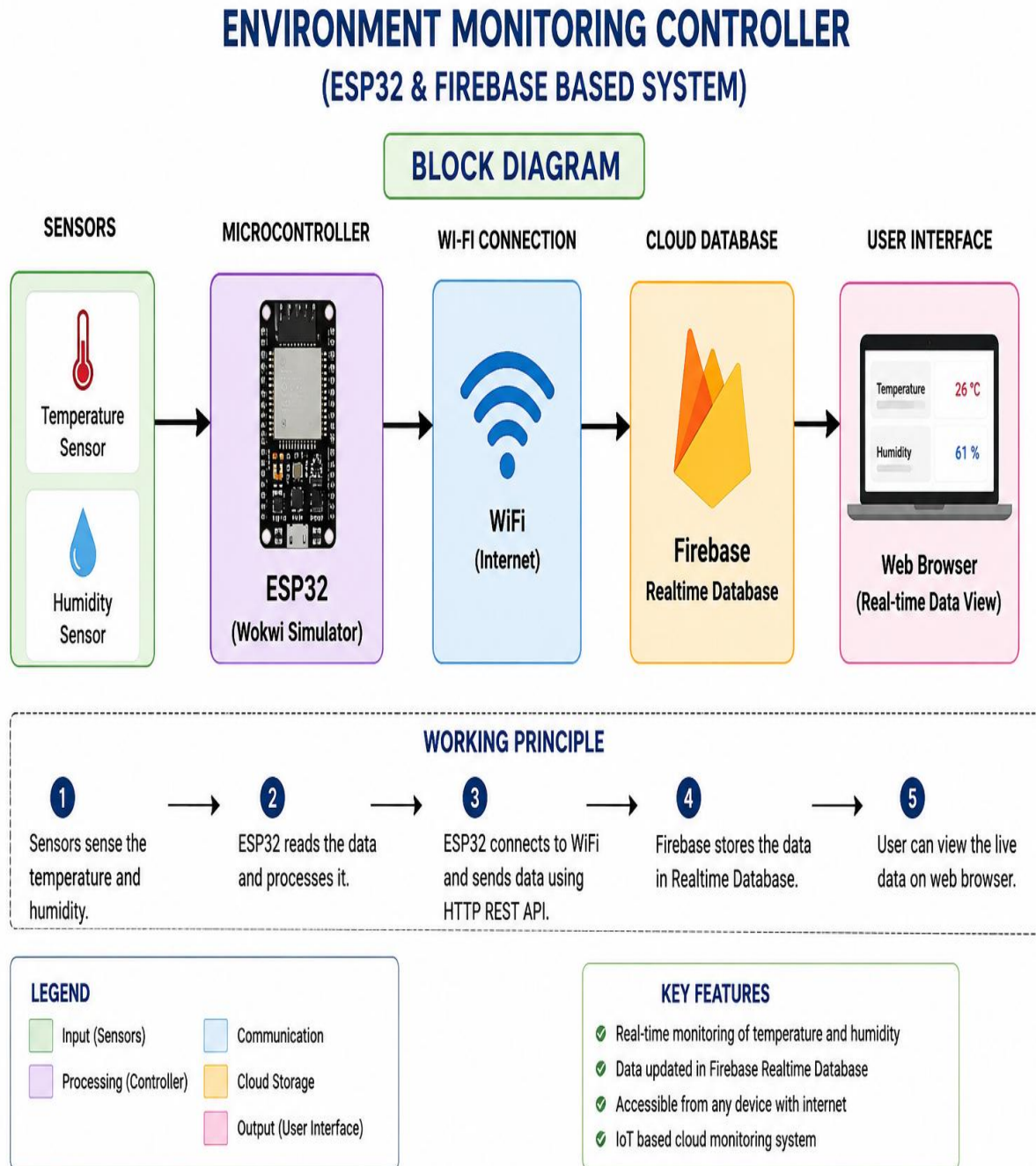
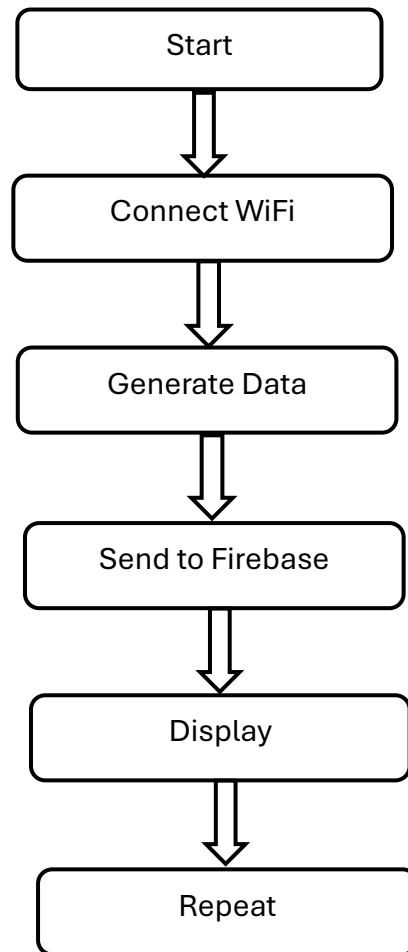


Fig 1.1 Block Diagram of Environment Monitoring Controller

6.Flow Chart



7.Implementation

7.1 Firebase Setup

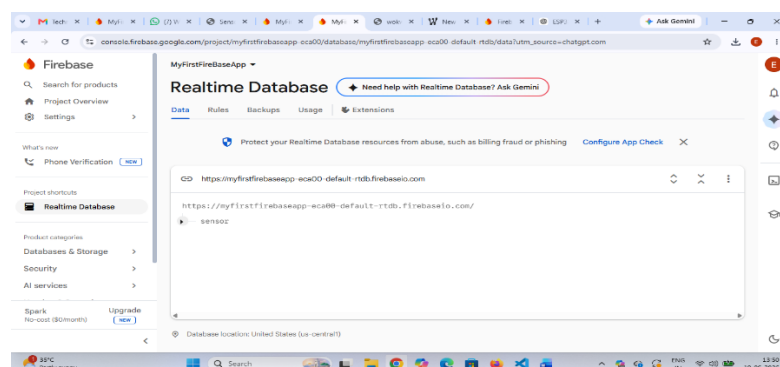


Fig 7.1 Firebase Project Setup

7.2 ESP32 Firebase Code(Wokwi)

```
#include <WiFi.h>

#include <HTTPClient.h>

#define WIFI_SSID "Wokwi-GUEST"

#define WIFI_PASSWORD ""

// Your Firebase URL
String firebaseHost = "https://myfirstfirebaseapp-eca00-default-rtdb.firebaseio.com";

void setup() {
  Serial.begin(115200);
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Connecting WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi Connected");
}

void loop() {
  // Only your required sensors
  float temp = 28 + random(-2, 3);
  float hum = 65 + random(-5, 5);
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;

    String url = firebaseHost + "/sensor.json";
    http.begin(url);
    http.addHeader("Content-Type", "application/json");

    // ONLY temperature & humidity
    String jsonData = "{";
    jsonData += "\"temperature\":" + String(temp) + ",";
```

```

jsonData += "\"humidity\": " + String(hum);
jsonData += "}";

int httpResponseCode = http.PUT(jsonData);

Serial.println("Updated Firebase:");

Serial.println(jsonData);

http.end();
}

delay(3000);
}

```

7.3 Output

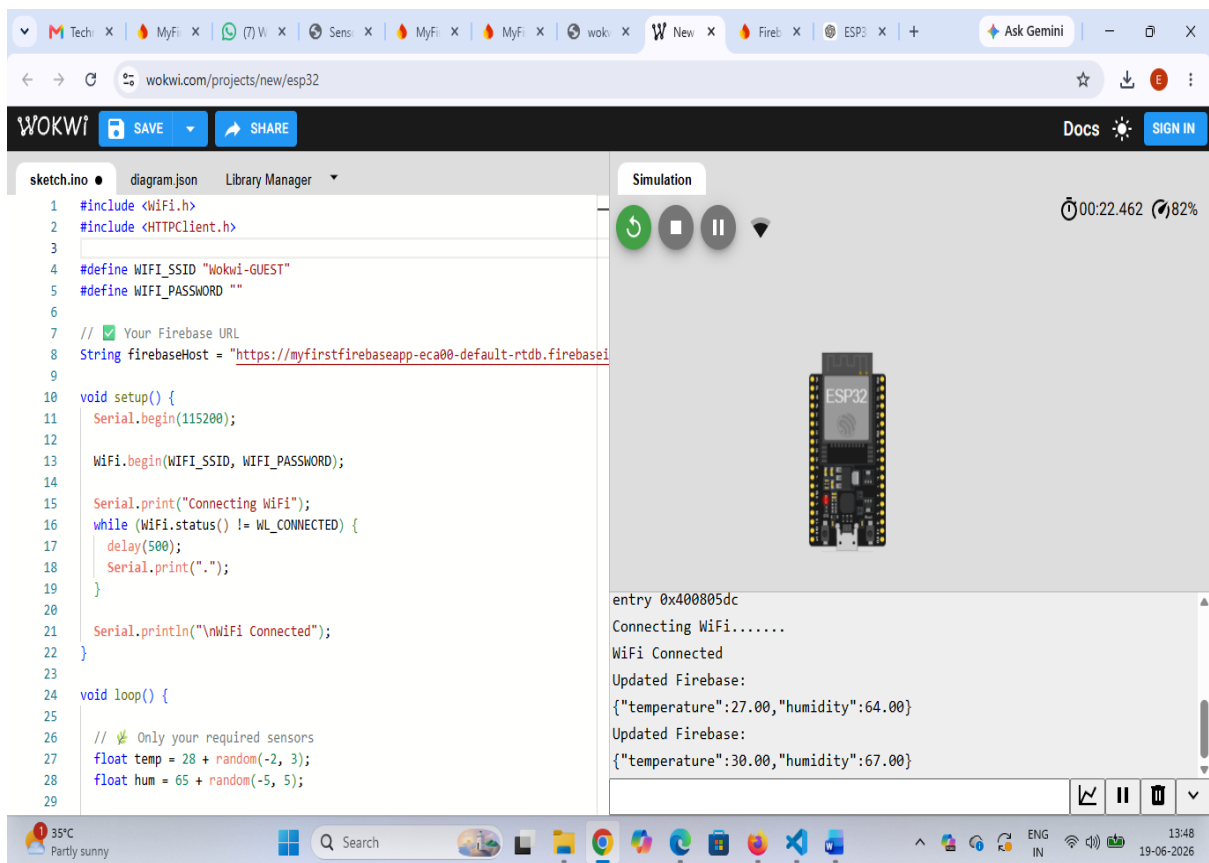


Fig 7.3.1 Simulation in Wokwi

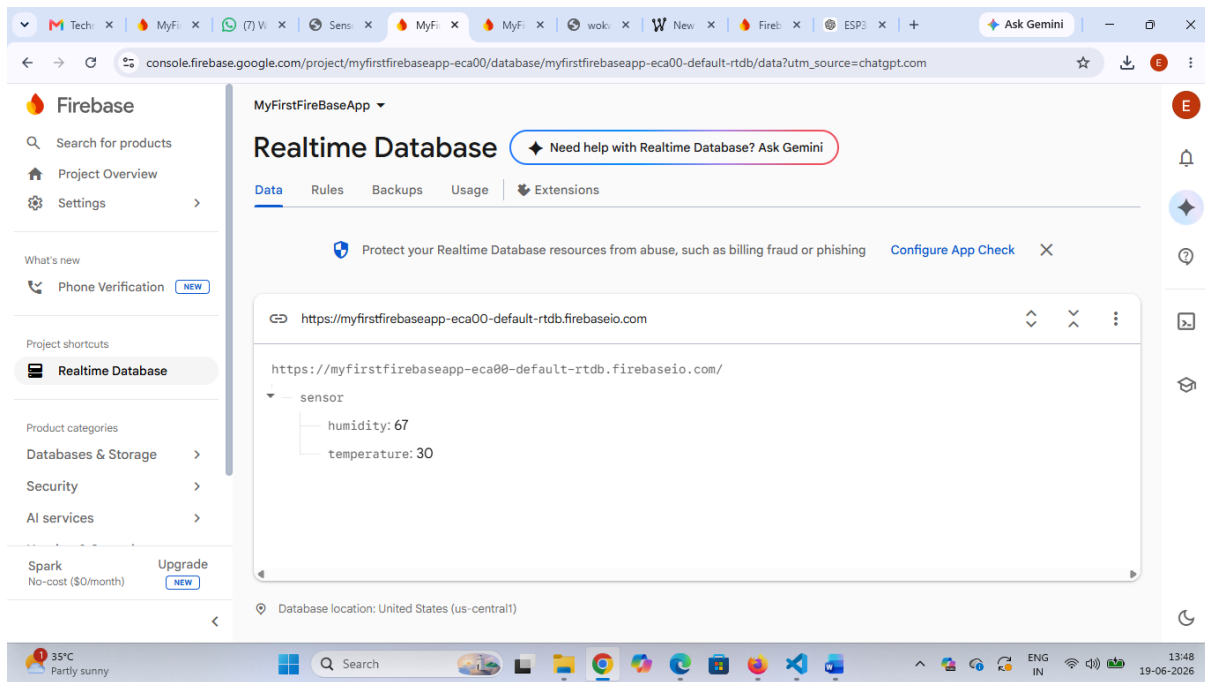


Fig 7.3.2 Output in Firebase

8.Conclusion

This project demonstrates a simple IoT-based real-time data monitoring system using ESP32 and Firebase Realtime Database. The system successfully collects sensor data (such as temperature and humidity) and transmits it to the cloud through WiFi using an HTTP REST API. The data stored in Firebase is updated in real time, allowing users to monitor the information remotely through a web browser. This improves accessibility, speed of monitoring, and reduces the need for manual data collection.

Overall, the project shows how embedded systems and cloud technology can be integrated to build efficient and scalable IoT applications. It also provides a strong foundation for advanced projects like smart home systems, environmental monitoring, and industrial automation.